

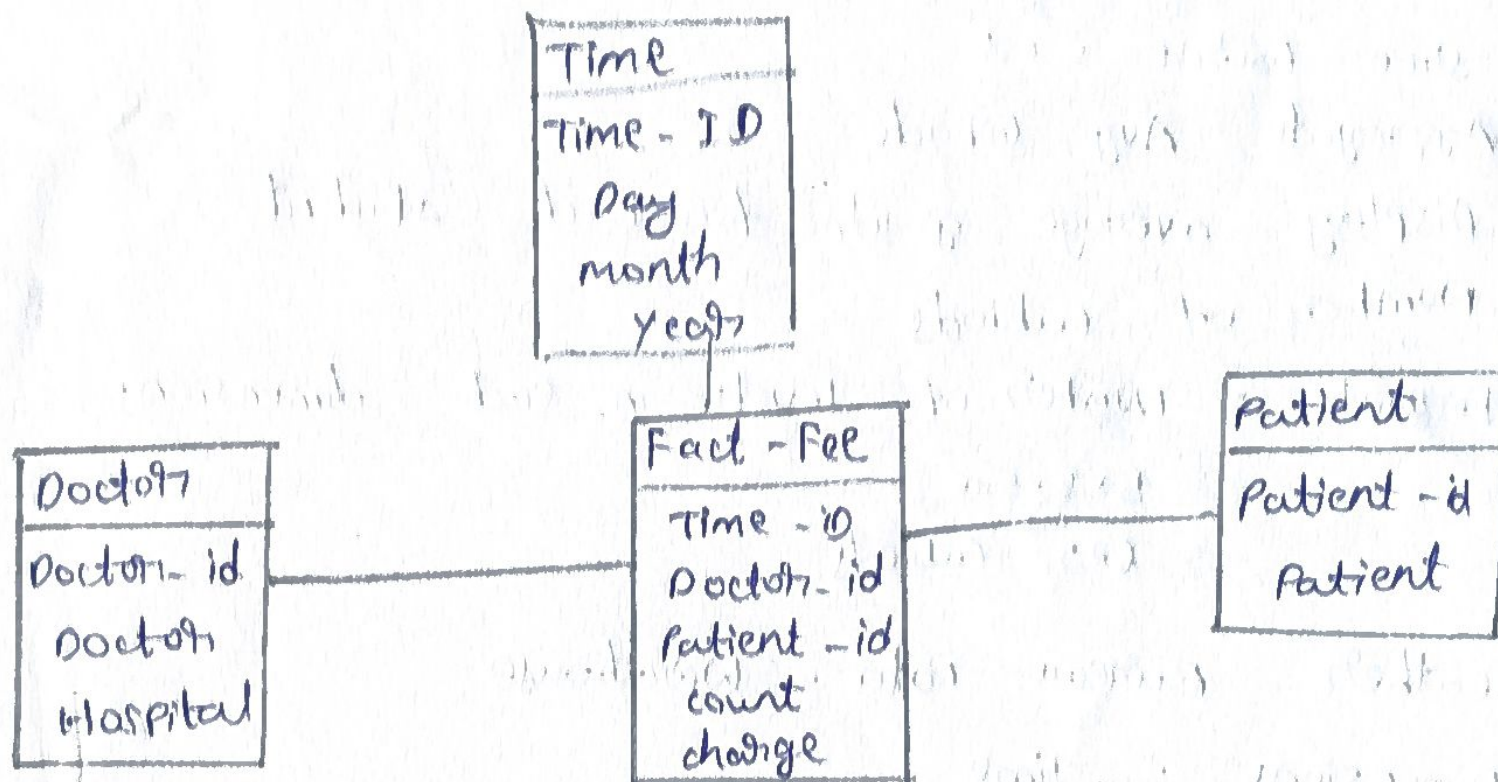
a) Three popular schemas used for data warehouses

* star schema

* snowflake schema

* Fact constellation (Galaxy) schema

b) star schema diagram



c) SQL query

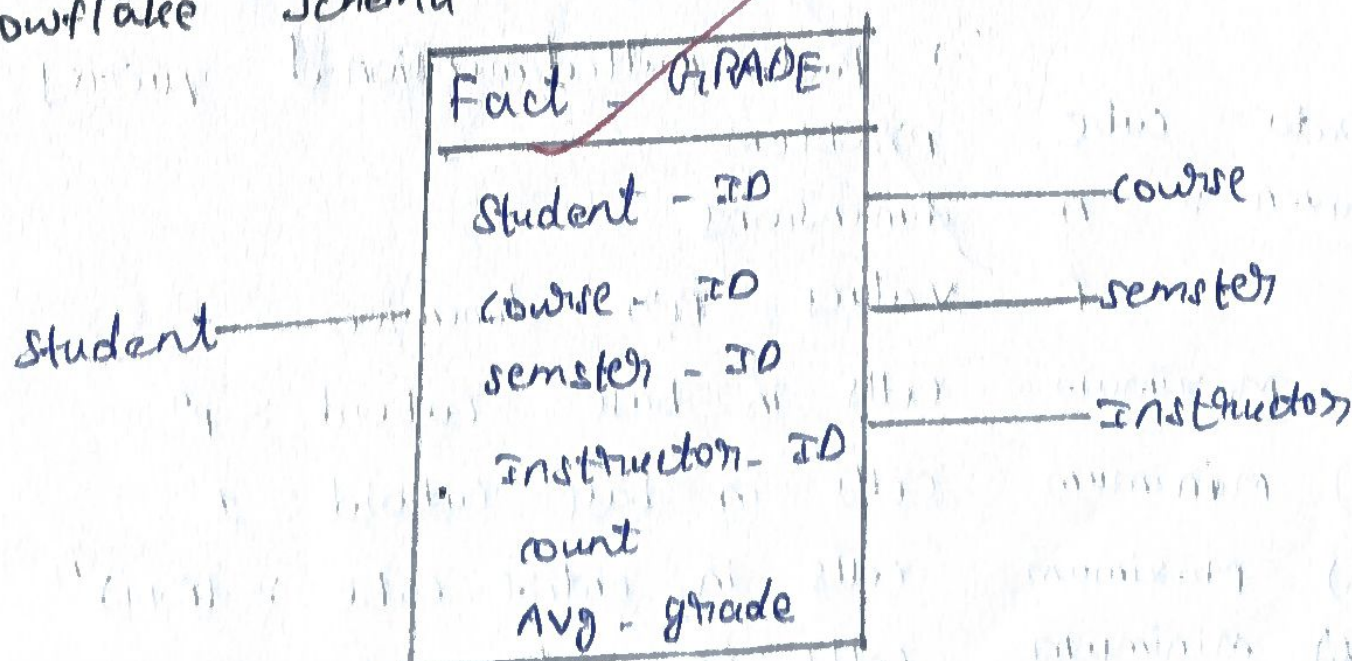
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SELECT      years, month, doctor, patient,
            SUM(count) as Total - visits,
            SUM(charge) as Total - charge
  
```

FROM Fee

GROUP BY years, month, doctor, patient;

2. a) snowflake schema



b.) OLAP operations:

Starting Cuboid:

[Student, course, semester, instructor]

Operations:-

- * Roll-up semester $\rightarrow$ year
- * slice course = CS
- * Aggregate Avg-Grade
- * Display average grade for each student

c.) Number of cuboids

Formula = Number of levels in each dimension

$$= 5 \times 5 \times 5 \times 5$$

$$= 625 \text{ cuboids}$$

3.) Weather Bureau Data Warehouse

Dimensions :- $\rightarrow$ Time
 $\rightarrow$ Location
 $\rightarrow$ Probe

Measure :- $\rightarrow$ Temperature
 $\rightarrow$ Air pressure
 $\rightarrow$ Precipitation

Advantage :- $\rightarrow$ Supports OLAP analysis
 $\rightarrow$ Identify weather trends
 $\rightarrow$ Fast multidimensional queries

4.) Data cube problem
Given :- n dimensions
 p values per dimension

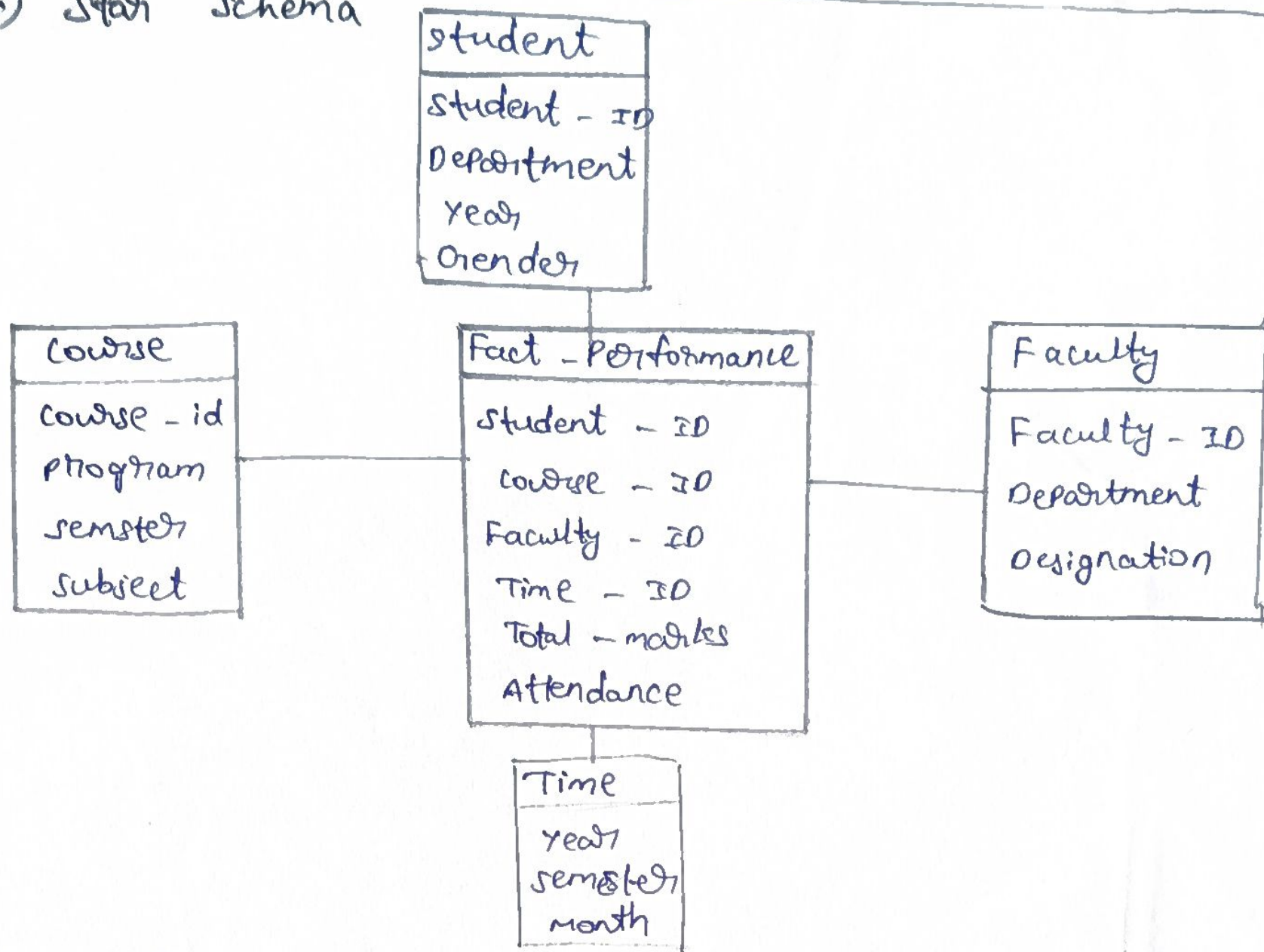
a.) Maximum cells in base cuboid :- p^n

b.) Minimum cells in base cuboid :- 1

c.) Maximum cells in entire cube :- $(p+1)^n$

d.) Minimum cells in entire cube :- 2^n

a.) Star schema



b.) OLAP operations

Base cuboid:- [student, course, Faculty, Time]

operations :- Roll up student $\rightarrow$ department
 * Roll up Time $\rightarrow$ Academic year
 * Aggregate Average marks

Result:- Department wise average marks for each academic year

c.) Number of cuboids

Levels

students = 4

course = 3

Faculty = 3

Time = 5

Formula :- $4 \times 3 \times 3 \times 5 = 180$ Cuboids